Tutorial 2

Ex. 8.4 Q47

① $\int \sqrt{x}\sqrt{1-x}\,dx = \int \sqrt{\underbrace{x-x^2}}\,dx$

CTS $\rightarrow b-(x+a)^2$

Substitute $(x+a) = \sin\theta$

② (Flav) Let $u = 1-x \Rightarrow du = -dx$

$$\underbrace{\int \sqrt{x}\sqrt{1-x}\,dx}_{0 \le x \le 1} = -\int \sqrt{1-u}\sqrt{u}\,du$$

$$= -\int \underbrace{\sqrt{u}\sqrt{1-u}\,du}$$

$$= -\int \sqrt{x}\sqrt{1-x}\,dx$$

$$2\int \sqrt{x}\sqrt{1-x}\,dx = 0 \Rightarrow \int \sqrt{x}\sqrt{1-x}\,dx = C \;??$$

Ex. 8.2 Q16

$$\int \underbrace{p^4}_{\text{diff}} \underbrace{e^{-p}}_{\text{int}} \, dp$$

$$= -p^4 e^{-p} - 4p^3 e^{-p} - 12p^2 e^{-p} - 24p e^{-p} - 24 e^{-p} + C$$

Tabular integration

	diff	int
$+$	p^4	e^{-p}
$-$	$4p^3$	$-e^{-p}$
$+$	$12p^2$	e^{-p}
$-$	$24p$	$-e^{-p}$
$+$	24	e^{-p}
$-$	0	$-e^{-p}$

Q21 $\int \underbrace{e^{\theta}}_{\text{int}} \underbrace{\sin\theta}_{\text{diff}} \, d\theta$

$$= e^{\theta}\sin\theta - e^{\theta}\cos\theta - \int e^{\theta}\sin\theta \, d\theta$$

	diff	int
$+$	$\sin\theta$	e^{θ}
$-$	$\cos\theta$	e^{θ}
$+$	$-\sin\theta$	e^{θ}

$\hookrightarrow -\int \sin\theta \, e^{\theta} \, d\theta$

Let $I := \int e^{\theta}\sin\theta \, d\theta$. Then

$$I = e^{\theta}\sin\theta - e^{\theta}\cos\theta - I$$

$$2I = e^{\theta}\sin\theta - e^{\theta}\cos\theta$$

$$I = \frac{1}{2}e^{\theta}\sin\theta - \frac{1}{2}e^{\theta}\cos\theta$$

Q41 $\int \underbrace{\sin 3x}_{\text{diff}} \underbrace{\cos 2x}_{\text{int}}\, dx$

① Product to Sum

② by Part (Do it now)

$= \frac{1}{2}\sin 3x \sin 2x$

$+ \frac{3}{4}\cos 3x \cos 2x$

$+ \int \frac{9}{4}\sin 3x \cos 2x\, dx$

	diff	int
+	$\sin 3x$	$\cos 2x$
−	$3\cos 3x$	$\frac{1}{2}\sin 2x$
+	$-9\sin 3x$	$-\frac{1}{4}\cos 2x$

$\int \frac{9}{4}\sin 3x \cos 2x\, dx$

Let $I := \int \sin 3x \cos 2x\, dx$

$\Rightarrow I = \text{———} + \frac{9}{4} I$

$I = \ldots$

$\int \sin 3x \cos 2x\, dx$

$= \frac{1}{2}\cos 2x \sin 3x - \int \frac{1}{2}\cos 2x \cdot 3\sin 3x\, dx$

$= \frac{1}{2}\cos 2x \sin 3x - \left(\qquad \right)$

Q56

$$\int \frac{(\sin^{-1}x)^2}{\sqrt{1-x^2}}\,dx$$

$u = \sin^{-1}x$

$du = \frac{1}{\sqrt{1-x^2}}\,dx$

$$= \int u^2\,du$$

$$= \frac{1}{3}u^3 + C$$

$$= \frac{1}{3}(\sin^{-1}x)^3 + C$$

$\sin^{\boxed{-1}}x \neq \frac{1}{\sin x}$

$$= \frac{1}{3}\sin^{-3}x + C$$ (Dont write like this)

Ex. 8.3

Product to Sum

$\downarrow$

$\sin(ax)\cos(bx)$

$a \neq b$

Q20 $\displaystyle\int_0^{\pi} 8\sin^4 y \cos^2 y \, dy$

(1) $\displaystyle = \int_0^{\pi} 8\sin^4 y \left(1 - \sin^2 y\right) dy$

$\displaystyle = \int_0^{\pi} 8\sin^4 y \, dy - \int_0^{\pi} 8\sin^6 y \, dy$

Conclusion: Not efficient

(2) $\displaystyle \int_0^{\pi} 8\left(\frac{1-\cos 2y}{2}\right)^4 \left(\frac{1+\cos 2y}{2}\right)^2 dy$

Binomial Theorem

Q30

$$\int_{\pi/2}^{3\pi/4} \sqrt{1-\sin 2x}\, dx \cdot \sqrt{\frac{1+\sin 2x}{1+\sin 2x}}$$

Conjugate

$$(1-\sin 2x)(1+\sin 2x) = 1-\sin^2 2x$$

$$1-\sin^2 2x = \cos^2 2x$$

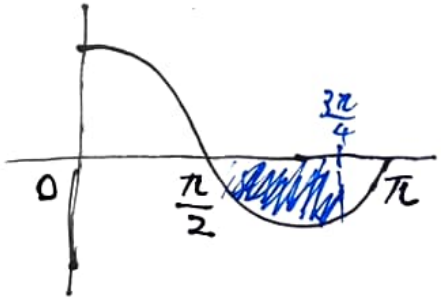
$$= \int_{\pi/2}^{3\pi/4} \frac{\sqrt{1-\sin^2 2x}}{\sqrt{1+\sin 2x}}\, dx$$

$$= \int_{\pi/2}^{3\pi/4} \frac{\sqrt{\cos^2 2x}}{\sqrt{1+\sin 2x}}\, dx$$

$$\sqrt{(f(x))^2} \neq f(x)$$

$$\sqrt{(f(x))^2} = |f(x)|$$

$$= \int_{\pi/2}^{3\pi/4} \frac{|\cos 2x|}{\sqrt{1+\sin 2x}}\, dx$$



$$= \int_{\pi/2}^{3\pi/4} -\frac{\cos 2x}{\sqrt{1+\sin 2x}}\, dx$$

$$u = 1+\sin 2x \Rightarrow du = 2\cos 2x\, dx$$

Q64 $\int \boxed{\sin\theta \cdot \sin 2\theta} \sin 3\theta \, d\theta$

$\sin A \sin B = \frac{1}{2}\cos(A-B) - \frac{1}{2}\cos(A+B)$

$= (\sim\cos__ - \cos__)\sin 3\theta \, d\theta$

~~sin cos~~ ~~sin~~

$= \sim\cos__\sin 3\theta + \sim\cos__\sin 3\theta$

PTS PTS

Ex. 8.4

Q 25

$$\int \sqrt{\frac{x+1}{1-x}}\,dx \cdot \sqrt{\frac{1+x}{1+x}}$$

① $u = \sqrt{\frac{x+1}{1-x}}$ (Tedious)

② Mult. by conj. $\Rightarrow \int \frac{|x+1|}{\sqrt{1-x^2}}\,dx \rightarrow$ $x = \sin\theta$.

$-1 \leq x \leq 1$

$\int \frac{x+1}{\sqrt{1-x^2}}\,dx$

$0 \leq x+1 \leq 2 \Rightarrow |x+1| = x+1$

always nonnegative